

Plasticity of temperature resistance in *Stegodyphus dumicola*

Bachelor's thesis (10 ECTS)

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Abstract

Climate change has drastically increased the interest in organisms' ability to respond to changes in their local environment, as species must migrate or adapt either through evolutionary adaptation or phenotypic plasticity to avoid extinction. Mostly species rely on a mixture of these responses, but less is known about how a species with limited genetic diversity manage to overcome climatic changes. The social spider species *Stegodyphus dumicola* was acclimated to five different temperature regimes and tested for critical thermal maximum and chill coma recovery. The species occupies a broad niche, but its genetic diversity is greatly reduced and therefore a plastic response was expected. The results showed a significant plastic response in heat tolerance, which suggests that species limited by their genetic diversity can occupy a broad climatic range due to phenotypic plasticity. The analysis showed no significant response in cold tolerance, which might be because the limited genetic diversity has fixed a less-plastic genotype or be due to an adaptation allowing survival in the lower thermal range. The results suggest that the broad thermal niche occupied by this species is facilitated by adaptive phenotypic plastic responses to heat tolerance, but further research in the species' cold acclimation abilities is needed.

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Introduction

Climate change is a global issue, which has brought with it a range of problems including changes in the mean temperatures in different regions of Earth and increased frequency of extreme thermal events (Schou et al. 2017a). Even in the most progressive Representative Concentration Pathways (RCP) the average temperature on the planet is expected to increase by 2°C by 2100 (Chrintz 2015), which makes more species experience temperatures close to or even beyond their physiological limits unless they are capable of responding to these changes (Araújo et al. 2013, Gunderson and Stillman 2015). Major shifts have been discovered in both geographic distribution and phenologies in a number of species from different taxa, which signifies that biological systems are responding to the current climate changes. Some organisms might find themselves incapable of responding properly to the local changes, and such organisms will be in great risk of going extinct (Chown et al. 2010, Hoffmann et al. 2013).

Populations inhabiting changing environments can respond in multiple ways as to avoid extinction; migration enables the species to search for a suitable habitat elsewhere, evolutionary adaptation allows the species to become better suited for its environment through genetic changes due to natural selection and adaptive plasticity allows a genotype to respond to different environments with different phenotypes. Successful evolutionary adaptation is known to depend on generation time and the genetic diversity in the population. The population size is often key to genetic diversity with greater population size increasing the potential genetic diversity. The genetic diversity in many populations today is decreased due to lacking gene flow between subpopulations (Hoffmann et al. 2015), as migration might not be possible due to habitat fragmentation and both natural and anthropogenic barriers too great to conquer

(Hoffmann et al. 2015). Besides being constrained by limited genetic diversity evolutionary adaptations are known to be slow, and may thus be untimely to secure the survival of the local population (Kristensen et al. 2008, 2015).

For organisms living in fragmented habitats with little genetic diversity phenotypic plasticity may allow survival in a climatic niche outside what had been expected to be possible. Plastic responses might be particularly important for organisms in changing environments to secure time for an evolutionary adaptation (Kristensen et al. 2015, Schou et al. 2017b). Adaptive plasticity can alter behaviour, morphology and physiology and can respond in all stages of life. A plastic response is induced by a genotype's ability to give immediate phenotypic responses to variation experienced in the environment that individual is experiencing. Some plastic responses are activated over months while others take only minutes to respond (Kristensen et al. 2016).

Phenotypic plasticity for thermal tolerances has been observed in both plants and animals generally showing a greater variance in cold tolerance than heat tolerance, owing to a potential upper physiological heat tolerance limit. For species already living near their physiological limit this loft can result in critical heat stress (Araújo et al. 2013, Hoffmann et al. 2013). Not all plastic responses are adaptive and can be limited by resources and the individual genotypes plastic extent, which often makes phenotypic plasticity a short-term solution, but these responses can be both reversible and non-reversible and might thus play a role in the survival of future generations by cross-generational effects (Hoffmann and Sgró 2011, Bijlsma and Loeschcke 2012, Schou et al. 2015).

To investigate how a species with limited genetic diversity might respond to environmental change the social spider

species *Stegodyphus dumicola* was chosen. The species' lack of genetic diversity is caused by a social inbreeding structure, where sibling-mated females create new nests, which is known as propagule dispersal. The propagules disperse by ballooning when local conditions deteriorate or the nest becomes too big, but successful establishment is rare due to high mortality rates for small nests. Due to this dispersal technique gene flow between nests are limited further decreased by the intra-family mating in the nest. Keeping the effective population size (N_e) high, inbreeding level low and the sex ratio close to 1:1 preserves genetic diversity in small, secluded populations (Chown et al. 2010, Settepani et al. 2017). Each nest of *Stegodyphus dumicola* instead experiences a reproductive skew with less than half off the females mating and a female-biased sex ratio up to 85% females, which further depletes the population of genetic diversity. This results in nests where the individuals are close to completely homozygous. Thus both evolutionary adaptation and migration are unlikely survival strategies for this species, which nevertheless occupies a wide niche with different climates (Settepani et al. 2014, 2017).

In most cases the selection on genetic diversity and phenotypic plasticity assist each other to best respond to a changing environment. Diversity is found in the plastic ability of one genotype and therefore selection might opt for a plastic genotype in changing environments resulting in increased fitness of the individuals possessing such a genotype, which makes evolutionary adaptation and adaptive plasticity difficult to separate (Kristensen et al. 2016, Schou et al. 2015). *Stegodyphus dumicola* is expected to adjust to changes primarily through phenotypic plasticity, and thus we acclimated the spiders to five different temperatures (15, 19, 23, 25 and 29° Celsius) and used thermal tolerance assays to examine, how they might adjust to the change in their local

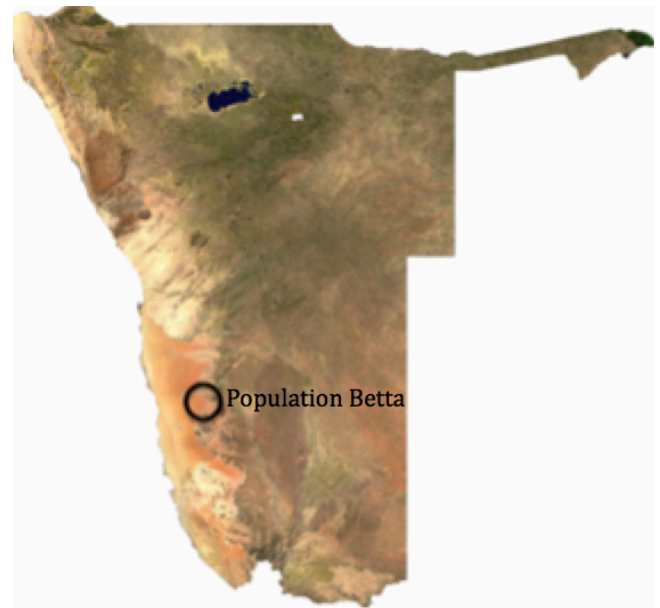


Figure 1: Collection site in Namibia for *Stegodyphus dumicola*. The circle marks the areas in which the spider nests were collected.

environment. Due to the limited genetic diversity of this species it is expected that all difference in temperature resistance following the acclimation can be accepted as a plastic response to the acclimation temperature.

Materials and methods

Experimental population

In this experiment an adult population of *Stegodyphus dumicola* was used due to the species' extremely low within-population diversity, with the lowest diversity estimated at $\pi_s = 0.00003$, and the homozygosity of the individuals occupying a nest (Settepani et al. 2017).

The experimental populations were collected in Namibia, where 15-20 spider nests were collected from each collection site. A nest refers to a group of adults and their offspring all originating from the founding propagule. The spiders used in this experiment originate from a collection site in the Namibian Hardap region titled 'Population

Betta'. Eighteen nests from 'Population Betta' were used in this experiment.

At Aarhus University each nest were kept at room temperature (21°C) and fed with flies and crickets. The spiders from each nest were distributed into boxes containing 30 spiders for each acclimation temperature. One such box will from hereon be referred to as a colony.

Acclimation process

For this experiment 90 colonies went through the acclimation process with 18 colonies at each of the 5 acclimation temperatures: 15°C, 19°C, 23°C, 25°C and 29°C. The 5 colonies originating from one nest were distributed to each of the acclimation temperatures.

The colonies were placed in acclimation rooms, where the temperature was kept constant over the course of 6 weeks. The acclimation rooms were sat on a permanent light/dark photoperiod with lights on from 08 to 20 (12h: 12h). During the acclimation process the spiders were fed on flies and crickets. After the acclimation process, each colony was split and the spiders placed in individual containers, weighed and numbered before being put in vials with corresponding numbers for the assay.

Thermal tolerance assay

From each colony 10 spiders were chosen at random for assessment of critical thermal maximum and another 10 for chill coma recovery. The spiders were placed in individual glass vials, which were sealed of, before testing the spiders' critical thermal maximum (CT_{max}). The spiders were randomly placed in a rack with the identification number visible. The rack were submerged in a 25°C water bath where the water temperature was increased by 0.1°C per minute. Video cameras were turned on then the water temperature had reached 43°C and filmed the spiders while the temperature increased to 55 °C. At this point all spiders were dead. The footage was analysed for the last time each spider moved – this instance in

time were used to calculate the knock down temperature for each spider.

The spiders tested for chill coma recovery (CC_recovery) were also placed in individual vials and randomly placed on a rack before being submerged in a water and 1,2-ethandiol mixture. The bath was kept at 0 °C for 2.5 hours, at which time the spiders had gone into a cold induced coma. Hereafter the temperature started increasing. The CC_recovery were filmed for 45 minutes in which time the water on average had reached a temperature of 22.09°C, and the footage were used to tell, when the spiders had recovered by analysing the footage for each spiders first movement.

From some colonies the numbers had dwindled through the acclimation process and a lower number of spiders were left for testing. The spiders from the lowest acclimation temperature (15°C) fared worse than the others, and some colonies left as little as 5 spiders for heat tolerance testing. In a single instant from the chill coma recovery assay, the combined effect of low spider number from the acclimation temperature of 15°C and drowned spiders, only two data points resulted from one colony. In any case where the number of spiders left after acclimation resulted in an unequal number for testing the extra spider were given to the CT_{max} assay. Spiders motionless through the tests were excluded from the dataset and so were the spiders that drowned due to the lids not closing properly.

Statistical analysis

The effects of the different acclimation temperature regimes on heat tolerance (CT_{max}) and chill coma recovery (CC_recovery) were investigated by using linear models fit with lm function in the R package lme4 (R Core Team 2017). The body mass of the spiders tested for CT_{max} and CC_recovery as well as the resulting values from the thermal tolerance assays were tested for normality. All data was normally distributed using $P < 0.05$. The model for CT_{max} was made with CT_{max} as

Table 1. Summary of acclimation groups

Acclimation group	Total #		Trait (°C)		Body mass (mg)	
	Trait	Body mass	Mean	± SD	Mean	± SD
CT _{max} 15°C	143	149	49.67	1.092	9.143	7.874
CT _{max} 19°C	165	172	50.31	0.937	8.572	4.294
CT _{max} 23°C	176	178	50.57	1.087	14.38	7.557
CT _{max} 25°C	173	178	50.65	1.15	15.93	7.903
CT _{max} 29°C	172	173	51.16	0.554	15.89	6.029
CC_recovery 15°C	120	144	2.162	1.098	10.24	8.593
CC_recovery 19°C	153	167	2.278	0.723	8.853	3.379
CC_recovery 23°C	162	177	2.355	0.644	15.03	7.368
CC_recovery 25°C	167	178	2.229	0.702	16.86	9.412
CC_recovery 29°C	168	172	2.267	0.707	16.52	5.961

Summary table containing values of: Total number of spiders surviving acclimation (Total # Body mass), data points after assays (Total # Trait), mean and standard deviation for both the CT_{max} assay, the CC_recovery assay and the spiders' body mass.

the response variable and acclimation temperature (Treatment), individual spider body mass (Body mass) and the interaction between the two as the explanatory variables. The CC_recovery analysis was conducted with the same explanatory variables as the previous test and with CC_recovery as the response variable. The spiders' body mass was included in the model to test for any influence on the resulting CT_{max} and CC_recovery. A one-way analysis of variance (Type I ANOVA) was conducted to make an F-test from which P-values were obtained for both linear models. The linear models and statistical analyses were performed in R.

Results

The results from the statistical analysis of the effect of acclimation on temperature resistance showed a significant plastic response to heat tolerance, as the spiders acclimated to higher temperatures fared better in the CT_{max} assay (Table 2) with a ~ 1.5 °C increase in heat tolerance for the spiders acclimated to 29°C compared to the spiders acclimated to 15 °C. The model suggests a positive linear relationship between acclimation temperature and CT_{max}. There were no significant effect of treatment

Table 2. Results from statistical analysis

Trait	Variable	Estimate	SE	F-test	P-value
CT _{max}	Intercept	48.434	0.315		
	Treatment	0.093	0.015	180.27	<0.001 *
	Body mass	-0.014	0.023	0.26	0.61
	Treatment*Body mass	0.001	0.001	0.254	0.615
CC_recovery	Intercept	2.562	0.263		
	Treatment	-0.010	0.012	0.65	0.42
	Body mass	-0.038	0.018	2.77	0.096
	Treatment*Body mass	0.001	0.001	3.285	0.070

The results were obtained from the statistical analysis of the change in heat tolerance (CT_{max}) and chill coma recovery (CC_recovery). CT_{max} and CC_recovery were analysed using a linear model. Type I anova provided the F-test for both traits from which the P-values were obtained. * P < 0.001.

and therefore no evidence of phenotypic plasticity in chill coma recovery. The body mass had no significant influence on neither CT_{max} nor $CC_{recovery}$, even though spiders acclimated to the two lowest temperatures ($15^{\circ}C$ and $19^{\circ}C$) were smaller than the spiders from the other acclimation temperatures. The spiders acclimated to $15^{\circ}C$ also fared worse in the acclimation process with a higher number of spiders dying during the acclimation (Table 1). The linear models also tested for any effect of the interaction between the acclimation treatment and the spiders' body mass and in both cases the models suggested the interaction had no influence.

Discussion

Organisms' ability to respond and potentially adapt to their environment is critical in understanding ecology, species distributions and ecological networks (Schou et al. 2017a). Here the plasticity of temperature resistance in *Stegodyphus dumicola* has been studied. The thermal tolerance assays for this species suggest, that plasticity is higher in heat tolerance than cold tolerance, which is contrary to findings in many other species (Araújo et al. 2013, Schou et al. 2017a, 2017b).

At Aarhus University the spiders were left at room temperature for a period of time until all 90 colonies had been prepared and the acclimation process were ready to begin. This might have resulted in a pre-testing acclimation of the spiders to $21^{\circ}C$. The spiders originate from a fluctuating environment where plastic abilities are expected to be beneficial, especially when the fluctuations are predictable. The pre-acclimation to a constant environment might have limited some of the plastic responses, as maintaining plasticity in constant environments is believed to be costly (Overgaard et al. 2011, Manenti et al. 2014). Plasticity can be calculated as acclimation response ratio,

which is "the slope of the line describing the change in thermal tolerance with a given change in acclimation temperature" (Gunderson and Stillman 2015). The estimated mean CT_{max} acclimation response ratio for terrestrial ectotherm taxa is estimated to 0.12 ± 0.01 (Gunderson and Stillman 2015). In the CT_{max} assay the filming started when the temperature of the water bath reached $43^{\circ}C$. This can potentially have resulted in bias in the data, as spiders dying before filming started were excluded from the data. Such data points could potentially increase the acclimation response for *Stegodyphus dumicola*, which in this study were estimated to $0.106^{\circ}C$ per centigrade increase in acclimation temperature.

The spiders assessed here originated from one of the cooler part of Namibia experiencing average monthly mean temperatures between $15^{\circ}C$ and $30^{\circ}C$ (<https://www.worldweatheronline.com/hardap-weather-averages/hardap/na.aspx>). Potentially the spiders are genetically adapted to these environmental conditions, as they have reached this part of Namibia through warmer surrounding areas. Plasticity in cold tolerance is more often associated with variation in thermal environments (Gunderson and Stillman 2015) and therefore the lack of plastic response in the $CC_{recovery}$ assay could be due to the chosen acclimation temperatures not exceeding the normal thermal mean range of the spiders. Although the poorer survival rate of the spiders acclimated at $15^{\circ}C$ might indicate that the spiders had reached their lower thermal limit for sustained survival.

The chill coma recovery assays tested at which temperature the spiders regain homeostasis and thereby were able to move. The recovery of homeostasis might be plastic, but the mechanism controlling the plasticity might be genetically constrained as in isofemale lines of *Drosophila simulans* (Manenti et al. 2014). The limited genetic diversity of *Stegodyphus dumicola* might thus

explain the lack of phenotypic response in the CC_recovery assay, as the propagule dispersal used by this species results in founder effects increasing the chance of fixing a less plastic genotype.

This study has mainly focused on the phenotypic plastic responses utilized by organisms to adapt to changes in their local environment. The option to behaviourally adapt to thermal changes is another plastic response known as behavioural thermoregulation and is believed to be another method for organisms to increase their chance of survival in changing environments (Hoffmann et al. 2013, Schou et al. 2017b). Behavioural thermoregulation can protect organisms from life-threatening thermal extremes. As an example are ants known to move nest sites in search of suitable thermal habitats (Diamond et al. 2012). Honeybees survive the extreme cold conditions in winter by cooperative thermoregulation, where the queen and brood are kept warm by the swarming of the worker bees (Ocko and Mahadevan 2014). The spiders acclimated to 15°C and 19°C ate less and were smaller than the spiders acclimated to the higher temperatures. The spiders might survive the cold winter temperatures they are naturally exposed to by a behavioural adaptation searching for and hiding in small, isolated spots to keep them protected.

The findings of this study suggest, that *Stegodyphus dumicola* is capable of occupying a wide niche due to adaptive phenotypic plasticity to heat tolerance, allowing it to adapt to upper thermal changes experienced in its local environment. The lack of plasticity in cold tolerance might indicate, that the species is range limited by its lower physiological limit, though it could be interesting to examine whether *Stegodyphus dumicola* has indeed adapted to a cooler environment by assessing spiders from this species living in other areas and DNA test them to study any genetic differences.

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